

Teacher:	Ray Joshua Balingkit	**Subject:**	Computer Science
Week:	3 (Term 1)	**Year:**	10-11
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will become familiar with the Python editing environment and the process of writing, saving, running and correcting a short program. They will use comments and meaningful identifiers to improve readability. Students will also develop their understanding of assignment, variables and basic data types, predicting how stored values change and distinguishing numeric values from text and Boolean values.

End-of-Week Assessment/Check: How will you know they got it?

Teacher questioning, discussion of worked examples, the completed student tasks and the final checks in both lessons. Check saved source code, successful output and one explained readability choice. Ask students to identify each type and explain the final score. Check that students can explain their reasoning and apply the skill without relying on the worked example.

LESSON SLOTS - MONDAY & FRIDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	7 September 2026	By the end of this lesson, students will be able to run and save a short Python program using readable code.	**H:** Show a program with an unclear name and unreadable layout. Invite students to explain their prediction before demonstrating the example. Use their responses to identify the idea that needs clarification. **ID:** Introduce the editor, running code, comments and meaningful identifiers. Explain the difference between editing source code and running it. Discuss meaningful filenames, indentation, comments and readable identifiers as ways of making a program easier to understand and maintain. **WD:** Write, save and run a greeting program; demonstrate a syntax error and its correction. Type a short greeting, save it as a Python file and show how the output differs from the code.	**Support:** Provide a short code template and a visual save-run checklist. Identify the editor and output areas on the display. Guide students through matching quotation marks, then remove the template for a second attempt. **Extension:** Students improve a deliberately unclear version without altering its output and justify each change. They distinguish a useful explanatory comment from a comment that merely repeats the code.



			Remove a quotation mark, read the error message and explain why correcting syntax is different from changing the intended greeting. **YD:** Create a named greeting program with an appropriate comment, then reopen and run the saved file. Students submit the saved source file and demonstrate that it runs after reopening. A partner reads the code and identifies its purpose, the output statement and the comment that helps explain it. **C:** Check saved source code, successful output and one explained readability choice. Ask students to support their answer with a specific example from their work, and record any misconception that needs a further demonstration.	
2	11 Septemb er 2026	By the end of this lesson, students will be able to use variables and distinguish integer float string and Boolean values.	**H:** Compare 12, 12.5, "12" and True. Invite students to explain their prediction before demonstrating the example. Use their responses to identify the idea that needs clarification. **ID:** Explain assignment, changing a value and the four basic data types. Explain that assignment stores a value under a name and that a later assignment replaces the previous value. Compare a numeric value with a string containing digits and discuss when a Boolean value is useful. **WD:** Predict and run assignments that update a player score. Set score to 10, add 5 and then double it. Ask students to predict each value before running the code. Use type() to inspect an integer, a	**Support:** Provide examples of the four data types and a partially completed value table. Use a labelled storage-box analogy for assignment and ask students to explain one update orally before coding it. **Extension:** Students investigate the difference between 12 and "12" in an expression. They predict and explain a type-related error and propose a suitable conversion without changing the meaning of the data.



			decimal value, a player name and a completion flag. **YD:** Create variables for a game player, score, time and completion flag; update the score and display values. Students produce a small table of variable names, initial values, types and final values. They explain why the player name is a string and show that updating the score does not change the other variables. **C:** Ask students to identify each type and explain the final score. Ask students to support their answer with a specific example from their work, and record any misconception that needs a further demonstration.	
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Pearson Edexcel International GCSE Computer Science Student Book: Chapters 5 and 6
- Teacher computer and classroom display
- Classroom computers with Python 3 editor and individual save folders
- Prepared algorithm, tracing or Python practice tasks aligned with the lesson objectives
- Whiteboard and markers
- Student notebooks and pencils
- Success criteria, test tables and teacher observation checklist



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?